

Review: Confidence Intervals & Hypothesis Tests

STAT 223 – Applied Analytics

Torin Quinlivan

March 31, 2025



Confidence Intervals

Confidence Interval

A **confidence interval** is a set of values that theoretically contain the population parameter a given proportion of the time when the experiment is performed many times.

A confidence interval estimates a parameter of the population (usually the mean, the proportion/success probability, or the variance: μ , p or σ^2).

Example #1: Enrollment Percentage

Example

One of the variables in the `crime` data file is the percent of children enrolled in school in 2000 (`enroll00`). Calculate the three intervals for this variable. Use a 95% level for each.

Confidence Interval:

```
interval(enroll00, type="confidence", level=0.95)
```

This tells us that the 95% confidence interval is from 90.473 to 93.021. In other words, we are 95% confident that the **population mean** is between 90.473 and 93.021, *inclusive*.

Example #1: Enrollment Percentage

Example

One of the variables in the `crime` data file is the percent of children enrolled in school in 2000 (`enroll00`). Calculate the three intervals for this variable. Use a 95% level for each.

Confidence Interval:

```
interval(enroll00, type="confidence", level=0.95)
```

This tells us that the 95% confidence interval is from 90.473 to 93.021. In other words, we are 95% confident that the **population mean** is between 90.473 and 93.021, *inclusive*.

Confidence Interval

A **confidence interval** is a set of values that theoretically contain the population parameter a given proportion of the time when the experiment is performed many times.

If (L, U) is a $(1 - \alpha) \times 100\%$ confidence interval for the parameter, then $(1 - \alpha) \times 100\%$ of the intervals – under repeated experiments – contain the parameter.

Definitions

Note that the confidence interval:

- gives some information about the population parameter.
- is a set of *reasonable* values for that parameter.
- is a function of the data (and thus a random variable).
- is a result of a probability distribution calculation.

Note that the confidence interval does *not* provide a probability statement on the parameter.

That is, one **cannot state** something like “There is a 95% *chance* that the parameter is between 4 and 9.”

Confidence Level

The **confidence level** is the theoretical frequency (or proportion) of possible confidence intervals that contain the true value of their corresponding parameter.

- The value of the confidence level is $c = 1 - \alpha$.
- By default, we will use $\alpha = 0.05$ in this course.
- The value α will be re-introduced when we cover hypothesis testing. It will represent the theoretical Type I error rate. It is selected by the research to reflect tradition in the discipline and the peculiarities of the current research.

The Z Theory

This is fine if all we have is the data. However, we know that sample means tend toward Normality as the sample size increases (CLT).

Thus, we may be able to create confidence intervals *without* bootstrapping and *with* using **additional information**.

Recall that for large n ,

$$\bar{X} \sim \text{NORMAL}\left(\mu, \frac{\sigma^2}{n}\right)$$

Applying the z -transform (i.e., standardizing formula, z -score formula) to $\bar{X}$ gives

$$\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim \text{NORMAL}(0, 1)$$

The Z Theory

$$\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim \text{NORMAL}(0, 1)$$

This means the expression on the left is between -1.96 and 1.96 approximately 95% of the time.

This is because the 2.5th percentile of the standard NORMAL distribution is -1.96 and the 97.5th percentile of the standard NORMAL distribution is 1.96 .

Thus, by definition, a theoretical 95% interval for $\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}}$ is between -1.96 and 1.96 .

The Z Theory

Since a theoretical 95% interval for $\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}}$ is between -1.96 and 1.96 , we can solve for μ to get a 95% confidence interval for μ , the population parameter.

One bound of a 95% confidence interval for μ is:

$$\begin{aligned}-1.96 &= \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \\ -1.96\sqrt{\frac{\sigma^2}{n}} &= \bar{X} - \mu \\ -\bar{X} - 1.96\sqrt{\frac{\sigma^2}{n}} &= -\mu \\ \bar{X} + 1.96\sqrt{\frac{\sigma^2}{n}} &= \mu\end{aligned}$$

The Z Theory

Similarly, the other bound of a 95% confidence interval for μ is:

$$1.96 = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}}$$

$$1.96\sqrt{\frac{\sigma^2}{n}} = \bar{X} - \mu$$

$$-\bar{X} + 1.96\sqrt{\frac{\sigma^2}{n}} = -\mu$$

$$\bar{X} - 1.96\sqrt{\frac{\sigma^2}{n}} = \mu$$

The Z Theory

Thus, the endpoints of a confidence interval for μ are

$$\bar{X} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma^2}{n}}.$$

Here, $z_{\frac{\alpha}{2}}$ is the absolute value of the $\alpha/2$ quantile for the $Z \sim \text{NORMAL}(0, 1)$ distribution. If $\alpha = 0.05$ as we usually use, then $z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$.

The Z Theory

This confidence interval for μ works if:

- You are estimating μ .
- You are able to calculate $\bar{x}$ and n .
- You know σ^2 .
- The data are generated from a NORMAL process (or n is sufficiently large).

The Z Theory

Recall the definition of the population variance. To calculate it, you need to know μ . However, since we are estimating μ in this procedure, we *rarely* know σ^2 .

Since we cannot use the z -procedure, we need to use a different procedure.

Going forward, all our procedures will have requirements to consider. Before you use any procedures, you need to consider if the requirements are reasonable to apply to your problem.

The t Theory

We rarely know the population variance, σ^2 , but we can always calculate the sample variance, s^2 . Can we use s^2 in place of σ^2 ?

Thanks to William Sealy Gossett and Karl Pearson in 1908, it can be show that the following is true:

$$\frac{\bar{X} - \mu}{\sqrt{s^2/n}} \sim t_\nu,$$

where $\nu = n - 1$ (in this one-sample case).

The t Theory

From this, we can derive the following confidence interval:

$$\bar{X} \pm T_{\frac{\alpha}{2}, \nu} \sqrt{\frac{s^2}{n}}$$

Here, $T_{\frac{\alpha}{2}, \nu}$ is the $\alpha/2$ quantile for the t distribution with $\nu = n - 1$ degrees of freedom.

For instance, if $\alpha = 0.05$ and $n = 4$, then $\nu = n - 1 = 3$ and $T_{\frac{\alpha}{2}, \nu} = t_{0.025, 3} = 3.182$.

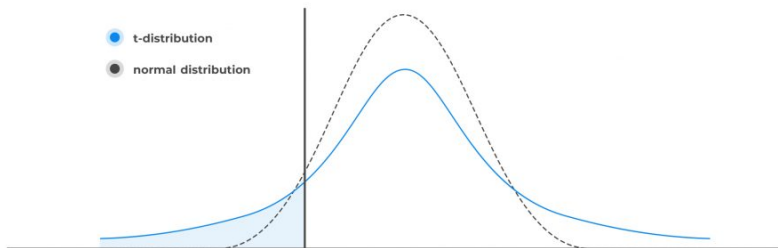
William Sealy Gossett



Student's t Distribution



T-distribution vs Normal Distribution



The Z Theory

This confidence interval for μ works if:

- You are estimating μ .
- You are able to calculate $\bar{x}$, n and s .
- The data are generated from a NORMAL process (or n is “large enough”).

Other Procedures

Note the process we followed to obtain the endpoints of the confidence intervals:

1. Create a test statistic.
2. Determine the distribution of that statistic.
3. Determine the quantiles of that distribution.
4. Solve the test statistic for the parameter of interest.

This process can/will be repeated for other parameters to create appropriate analysis procedures.

You will use it many times in MATH 322: Mathematical Statistics II.

The Theory Behind Hypothesis Testing

The theory behind hypothesis testing is:

1. State the research hypothesis and the null hypothesis.
2. Determine how much the data support the null hypothesis:
 - Determine the parameter tested.
 - Determine appropriate statistic.
 - Determine distribution of that statistic assuming the null hypothesis is true.
 - Determine how likely it is to observe the statistic (data) if the null hypothesis is true.
3. Interpret that level of support.

Definition: Hypothesis

Hypothesis

A **hypothesis** is a testable claim about reality.

Since it is a claim about reality, it concerns some aspect of the population (a parameter). The usual parameters hypothesized about at this level are the mean μ , proportion p , and variance σ^2 . But we can hypothesize about any aspect of the population.

The “generic” parameter is θ .

Since it is a claim about reality, it separates all possible realities into those that are consistent with the hypothesis and those that are incompatible with it.

Definition: Research Hypothesis

The most important hypothesis is the one made by the researcher:

Research Hypothesis

A **research hypothesis** is a testable claim about reality *made by the scientist*.

This is the claim that the statistician must eventually come to a conclusion about.

Because we are using statistics to test this hypothesis, we create two statistics-specific hypotheses. These two hypotheses divide all possible realities into two groups, those that are compatible with the research hypothesis and those that are not.

Definition: Null Hypothesis

Null Hypothesis

The **null hypothesis**, H_0 , is a statement of the status quo or our prior beliefs. It is what we assume is true unless we see evidence to disprove it.

The null hypothesis is always in some way a statement of equality.

Definition: Alternative Hypothesis

Alternative Hypothesis

The **alternative hypothesis**, H_A , is what we are trying to prove with our sample.

The alternative hypothesis is the opposite of the null hypothesis, and if we reject the null hypothesis, we accept the sample as evidence in support of the alternative hypothesis.

Comparing the Hypotheses

Below is a table of all possible research hypotheses and their corresponding null and alternative hypotheses:

H_R	H_0	H_A
$\theta < \theta_0$	$\theta \geq \theta_0$	$\theta < \theta_0$
$\theta = \theta_0$	$\theta = \theta_0$	$\theta \neq \theta_0$
$\theta > \theta_0$	$\theta \leq \theta_0$	$\theta > \theta_0$
$\theta \leq \theta_0$	$\theta \leq \theta_0$	$\theta > \theta_0$
$\theta \neq \theta_0$	$\theta = \theta_0$	$\theta \neq \theta_0$
$\theta \geq \theta_0$	$\theta \geq \theta_0$	$\theta < \theta_0$

The symbol θ is a generic symbol for a parameter. It can represent μ , p , σ^2 , or *any other parameter you can imagine*.

The symbol θ_0 represents the value claimed by the researcher. It is a number.

Comparing the Hypotheses

H_R	H_0	H_A
$\theta < \theta_0$	$\theta \geq \theta_0$	$\theta < \theta_0$
$\theta = \theta_0$	$\theta = \theta_0$	$\theta \neq \theta_0$
$\theta > \theta_0$	$\theta \leq \theta_0$	$\theta > \theta_0$
$\theta \leq \theta_0$	$\theta \leq \theta_0$	$\theta > \theta_0$
$\theta \neq \theta_0$	$\theta = \theta_0$	$\theta \neq \theta_0$
$\theta \geq \theta_0$	$\theta \geq \theta_0$	$\theta < \theta_0$

DO NOT MEMORIZE THIS TABLE! Instead learn what it says about the *relationships* between the research, null and alternative hypotheses.

What do all of the null hypotheses have in common? Why would that be a requirement? What is the relationship between the null and alternative hypotheses? Why does this make sense?

Definition: The p -value

p -value

The p -**value** is the probability of observing a sample with data this extreme, or more so, *given that the null hypothesis is true*.

p -value

The p -**value** is the probability of observing a test statistic this extreme, or more so, *given that the null hypothesis is true*.

p -value

The p -**value** is a measure of the amount of support in the data for the null hypothesis.

Definition: Decision Theory

We now have two of the three parts of the “hypothesis-testing theory” laid out earlier. The last part is making a decision about the research hypothesis based on the data.

Do the data sufficiently support the research hypothesis?

Note that the statement is binary: “Yes” or “No”.

Since the decision is binary, we need to determine a “cut-off” point between what sufficiently supports the research hypothesis and what does not. This threshold is referred to as “the alpha-value”, α .

If the level of support is less than α , we ***reject*** the null hypothesis. Otherwise, we do not, i.e. we ***fail to reject***.

Definition: Decision Theory

α

The **alpha value**, α is the Type I error rate claimed by the statistician.

Type I error

A **Type I error** occurs when the researcher rejects a true null hypothesis. This is also known as a *false positive*.

Type I error rate

The **Type I error rate** is the proportion of the time the researcher commits a Type I error (rejects a true null hypothesis).

Note: The actual Type I error rate may *not* actually be α , what it is claimed to be.

Definition: Decision Theory

Type II error

A **Type II error** occurs when the researcher fails to reject a false null hypothesis. This is also known as a *false negative*. It is symbolized by β .

Both α and β are error rates. We would like to minimize both.

- However, decreasing α results in increasing β .
- Furthermore, reducing either to zero results in the other being 100%.
- Fisher suggested that we specify α and let β be whatever it is.

Student's t -Test

If we do not know the population standard deviation, but can assume that the data are from a NORMAL process, we can perform the t -test, one of the most commonly used statistical tests.

The t -test assumes that, under the null hypothesis H_0 , the *test statistic* follows a t distribution with $df = n - 1$ degrees of freedom.

$$T = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$$

Then, we reject the null hypothesis if:

- $T \leq t_{0.05, n-1}$ if $H_A : \mu < \mu_0$
- $T \geq t_{0.05, n-1}$ if $H_A : \mu > \mu_0$
- $|T| \geq t_{0.05, n-1}$ if $H_A : \mu \neq \mu_0$

We can get the value for $t_{0.05, n-1}$ or the p -value from a t -table.

One-Parameter Procedures: μ

Parametric Procedure: Student's t -test

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from NORMAL distribution
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x)`
- R function: `t.test(x)`

One-Parameter Procedures: μ (and $\tilde{\mu}$)

Non-Parametric Procedure #1: Wilcoxon procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from symmetric distribution
 - Requirement test: Hildebrand Rule
 - `hildebrand.rule(x)`
- R function: `wilcox.test(x, conf.int=TRUE)`

One-Parameter Procedures: μ

Non-Parametric Procedure #2: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing!
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

One-Parameter Procedures: p

Parametric Procedure: Binomial procedure

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Data generated from BINOMIAL distribution
- R function: `binom.test(x, n)`

One-Parameter Procedures: p

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Nothing
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

This code assumes the data, `y`, consist of 0 and 1 values.

One-Parameter Procedures: σ^2

Parametric Procedure: Chi-Square (χ^2) procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from NORMAL process
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x)`
- R function: `onevar.test(x)`

One-Parameter Procedures: σ^2

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = var(x)
}

quantile(st, c(0.025, 0.975))
```

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Parametric Procedure: two-sample t -procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from NORMAL distribution – in *each* population
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x ~ g)`
- R function: `t.test(x ~ g)`
- R function: `t.test(x1, x2)`

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Non-Parametric Procedure: Mann-Whitney procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Beyond the scope of this course
- R function: `wilcox.test(x ~ g, conf.int=TRUE)`
- R function: `wilcox.test(x1, x2, conf.int=TRUE)`

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Dependent)

Parametric Procedure: two-sample dependent procedure

- This is just a one-sample procedure if you focus on the differences, $d = \mu_1 - \mu_2$, instead of the measurements.
- As such, refer to the one-sample mean procedure.
- This should help to illustrate the need to understand the data and what you want to understand.

Two-Parameter Procedures: $p_1 - p_2$

Parametric Procedure: Proportions procedure

- Graphic: Binomial plot (`binom.plot(x=c(x1, x2), n=c(n1, n2))`)
- Requires: Expected number of successes is at least 5 in each group
- R function: `prop.test(x=c(x1, x2), n=c(n1, n2))`

Two-Parameter Procedures: σ_1^2/σ_2^2

Parametric Procedure: Fisher's F -test

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from NORMAL distribution – in *each* population
 - Requirement test: Shapiro-Wilk test
 - R function: `shapiroTest(x ~ g)`
- R function: `var.test(x ~ g)`
- R function: `var.test(x1, x2)`